



Fix by Swapping

A block points the wrong way

Name:

Date:

★ I can fix a block that points the wrong way.

Wrong way!

Sometimes a block points the **WRONG** way — a back block where it should be forward, or a forward block where it should be back. Run the code, find the wrong-way block, and **SWAP** it to fix the program.

Bit's number path



Program 1 — target is 4

start at 1

forward 2

back 1

1 Run it, find the wrong way

Run Program 1 from 1. Where does Bit land? It misses 4. Circle the block that points the **WRONG** way (it moves Bit away from 4).

2 Swap it

Change the wrong-way block so Bit reaches 4. What should it be? Run the fixed code — what number now?

Program 2 — target is 1

start at 3

forward 1

back 1

3 You try — swap to fix

Run Program 2 from 3. It misses 1. Circle the wrong-way block, swap it, and run again. What should the block be, and where does Bit land?

Big idea

If a block sends Bit the wrong way, swap its direction. Forward becomes back, or back becomes forward — then run it again to check.



Fix by Swapping

Quick facilitation notes & answer key

What this worksheet teaches

Students fix a bug where one block points the wrong way by swapping its direction (forward to back, or back to forward), then run the fixed program to confirm it reaches the target. Changing a block to fix the result is the core of debugging. No coding experience needed.

Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: tape a number line 0 to 5 on the floor and mark the target, so students can walk the buggy program and feel Bit miss.

How to lead it (about 20 minutes)

1. Show them (you do). Run a program that misses the target and find the block that moves Bit the wrong way.
2. Do one together. Exercise 1 — the wrong-way block in Program 1 is back 1.
3. Let them try. Students swap it and re-run (Ex 2), then fix Program 2 (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Program 1: 1, forward 2 → 3, back 1 → 2. The wrong-way block is back 1.

Exercise 2 — swap back 1 to forward 1: 1 → 3 → 4. Bit reaches the target, 4.

Exercise 3 — Program 2: 3, forward 1 → 4, back 1 → 3 (misses 1). Swap forward 1 to back 1: 3 → 2 → 1. Bit reaches the target, 1.

If students get stuck — and ways to adjust

Watch for: changing the number instead of the direction. Here the fix is to flip forward to back (or back to forward).

Make it easier: walk the program and notice the step that goes away from the target.

Make it harder: ask which way Bit would land if they left the block as it was.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a program, change a block to fix it, and explain what the change did.

You'll know it worked when

The child swaps the wrong-way block and the fixed program reaches the target.